

MPHASIS ASE

100-Question First-Round Practice Set

Quantitative Aptitude • Logical Reasoning • Verbal Ability • Technical MCQ & CS Fundamentals

Purpose: A placement-style practice set built from your preparation master-sheet, the uploaded 95-question assessment, and your SQL/sorting MCQ material. CS fundamentals are added to broaden the technical section.

Questions	Sections	Answer Format
100 MCQs	25 Quant + 25 Logical + 20 Verbal + 30 Technical	4 options + correct option shown beside every question

How to use this PDF

- Do each section under a hard timer. The uploaded mock uses a sectional format with time limits.
- Because the correct option is intentionally visible, use this as a learning/revision set rather than a blind mock.
- For a real mock attempt, cover the green answer labels and reveal them only after solving.
- Technical practice emphasizes SQL, OOP, sorting, DBMS, OS, networking, and core data structures.

Section 1 — Quantitative Aptitude

25 questions • Correct option is displayed on the right of each question

Q01 A value is increased by 20% and then decreased by 10%. What is the net percentage change?

Correct: A

- A) 8% increase
- B) 10% increase
- C) 8% decrease
- D) 12% increase

Q02 A and B together finish a job in 12 days, B and C in 15 days, and A and C in 20 days. How many days would A alone take?

Correct: B

- A) 24
- B) 30
- C) 36
- D) 40

Q03 A car travels 180 km in 3 hours and then 120 km in 2 hours. What is its average speed for the whole journey?

Correct: C

- A) 50 km/h
- B) 55 km/h
- C) 60 km/h
- D) 65 km/h

Q04 A sum is divided into two parts in the ratio 3:5. The parts earn simple interest at 6% and 8% p.a. for 2 years. If total interest is ₹2,320, what is the sum?

Correct: C

- A) ₹14,500
- B) ₹15,000
- C) ₹16,000
- D) ₹17,400

Q05 If $x:y = 4:7$ and $y:z = 14:15$, then $x:z$ is:

Correct: B

- A) 4:15
- B) 8:15
- C) 14:15
- D) 8:21

Q06 The average of 8 numbers is 24. If one number 38 is replaced by 22, what is the new average?

Correct: B

- A) 21
- B) 22
- C) 23
- D) 24

Q07 The HCF of two numbers is 12 and their LCM is 420. If one number is 60, the other is:

Correct: B

A) 72
B) 84
C) 96
D) 108

Q08 What is the smallest four-digit number divisible by 19 whose digits add up to 19?	Correct: B
A) 1045	
B) 1216	
C) 1387	
D) 1558	

Q09 A and B invest ₹40,000 and ₹60,000 for one year. If the total profit is ₹25,000, B's share is:	Correct: C
A) ₹10,000	
B) ₹12,500	
C) ₹15,000	
D) ₹16,000	

Q10 The present ages of X and Y are in the ratio 4:7. If X is 12 years younger than Y, X's age is:	Correct: B
A) 12	
B) 16	
C) 18	
D) 20	

Q11 A boat travels downstream at 12 km/h and upstream at 8 km/h. Its speed in still water is:	Correct: C
A) 8 km/h	
B) 9 km/h	
C) 10 km/h	
D) 12 km/h	

Q12 Two fair dice are rolled. What is the probability that the sum is 9?	Correct: A
A) 1/9	
B) 1/12	
C) 1/6	
D) 5/36	

Q13 In how many ways can 5 distinct gifts be distributed among 3 children so that every child gets at least one gift?	Correct: B
A) 120	
B) 150	
C) 180	
D) 243	

Q14 A population of 40,000 grows by 10% per year, then 2,000 people migrate out at the end of each year. Population after 2 years is:	Correct: B
A) 44,000	
B) 44,200	
C) 44,800	
D) 45,200	

Q15 A tank can be filled by A in 12 minutes and emptied by B in 18 minutes. If both are opened together, the tank fills in:	Correct: C
A) 24 min	
B) 30 min	
C) 36 min	
D) 40 min	

Q16 An article costing ₹800 is sold at a profit of 15%. Its selling price is:	Correct: C
A) ₹880	
B) ₹900	
C) ₹920	
D) ₹940	

Q17 If $3x + 5 = 26$, then x is:	Correct: B
A) 5	
B) 6	
C) 7	
D) 8	

Q18 A clock gains 5 minutes every hour. If set correctly at 8:00 AM, what will it show at the correct time 2:00 PM?	Correct: C
A) 2:05 PM	
B) 2:20 PM	
C) 2:30 PM	
D) 2:35 PM	

Q19 A mixture contains milk and water in the ratio 5:2. If 14 litres of water are present, how much milk is present?	Correct: C
A) 25 L	
B) 30 L	
C) 35 L	
D) 40 L	

Q20 What is the remainder when 7^{100} is divided by 6?	Correct: B
A) 0	
B) 1	
C) 3	
D) 5	

Q21 A train 180 m long crosses a pole in 9 seconds. Its speed is:	Correct: B
A) 60 km/h	
B) 72 km/h	
C) 80 km/h	
D) 90 km/h	

Q22 The compound interest on ₹10,000 at 10% p.a. for 2 years, compounded annually, is:	Correct: B
A) ₹2,000	
B) ₹2,100	
C) ₹2,200	
D) ₹2,500	

Q23 12 workers finish a job in 24 days. If 16 equally efficient workers work on it, the job takes:	Correct: B
A) 16 days	
B) 18 days	
C) 20 days	
D) 22 days	

Q24 The area of a rectangle is 240 cm ² and its length is 20 cm. Its perimeter is:	Correct: C
A) 44 cm	
B) 48 cm	
C) 64 cm	
D) 72 cm	

Q25 A number is increased by 25% and becomes 250. The original number was:	Correct: B
A) 180	
B) 200	
C) 210	
D) 225	

Section 2 — Logical Reasoning

25 questions • Correct option is displayed on the right of each question

Q26 Find the missing number: 3, 8, 15, 24, 35, ?	Correct: B
A) 46	
B) 48	
C) 49	
D) 50	
Q27 Seven friends P, Q, R, S, T, U and V sit in a row. P is third from the left, Q is third from the right, and R sits exactly between P and Q. Who sits in the middle?	Correct: C
A) P	
B) Q	
C) R	
D) S	
Q28 A person walks 5 km north, then 12 km east. What is the shortest distance from the starting point?	Correct: A
A) 13 km	
B) 17 km	
C) 7 km	
D) 12 km	
Q29 Ravi is the brother of Meena. Meena is the mother of Arun. How is Ravi related to Arun?	Correct: B
A) Father	
B) Uncle	
C) Brother	
D) Grandfather	
Q30 If A=1, B=2, ..., Z=26, what is the value of CODE?	Correct: A
A) 27	
B) 29	
C) 31	
D) 33	
Q31 Statements: All pens are tools. Some tools are blue. Conclusions: I. Some pens are blue. II. All pens are tools. Which follows?	Correct: B
A) Only I	
B) Only II	
C) Both I and II	
D) Neither	
Q32 What is X's present age? I: $X:Y = 4:7$. II: X is 12 years younger than Y.	Correct: C
A) I alone sufficient	

- B) II alone sufficient
- C) Both together sufficient
- D) Either alone sufficient

Q33 In a class, R is 12th from the top and 18th from the bottom. How many students are there?

Correct: B

- A) 29
- B) 30
- C) 31
- D) 32

Q34 If 1 January 2026 is Thursday, what day is 15 January 2026?

Correct: B

- A) Wednesday
- B) Thursday
- C) Friday
- D) Saturday

Q35 A clock shows 2:20. What time will its mirror image show?

Correct: A

- A) 9:40
- B) 10:40
- C) 8:40
- D) 9:20

Q36 Find the next term: 2, 6, 12, 20, 30, ?

Correct: B

- A) 40
- B) 42
- C) 44
- D) 46

Q37 Which one is different from the others?

Correct: C

- A) Triangle
- B) Square
- C) Circle
- D) Rectangle

Q38 BOOK : READ :: FOOD : ?

Correct: B

- A) Cook
- B) Eat
- C) Buy
- D) Serve

Q39 Five people A, B, C, D and E sit in a row. A is left of B, C is right of B, D is left of A, and E is right of C. Who is in the middle?

Correct: B

- A) A
- B) B

C) C

D) D

Q40 G is the father of H. H is the sister of J. J is the father of K. How is G related to K?

Correct: B

A) Father

B) Grandfather

C) Uncle

D) Brother

Q41 A man faces east. He turns left, walks 10 m, turns left and walks 5 m, then turns right and walks 5 m. Which direction is he facing now?

Correct: A

A) North

B) South

C) East

D) West

Q42 If CAT is coded as DBU, how is DOG coded using the same rule?

Correct: A

A) EPH

B) EOG

C) DPH

D) FPI

Q43 Six friends A, B, C, D, E, F sit in a row. E is at the left end. B is immediately right of E. D is immediately right of C. There are two people between B and F. Who can be at the right end?

Correct: A

A) A

B) C

C) D

D) F

Q44 If P is taller than Q, Q taller than R, and S taller than P, who is the shortest?

Correct: C

A) P

B) Q

C) R

D) S

Q45 If the letters of MARKET are arranged alphabetically, which letter is fourth?

Correct: B

A) K

B) M

C) R

D) T

Q46 Find the missing term: 5, 11, 23, 47, ?

Correct: C

A) 93

- B) 94
- C) 95
- D) 96

Q47 A statement says: 'The company should provide laptops to all interns to improve productivity.' Which assumption is most relevant?

Correct: A

- A) Laptops can support interns' work
- B) All interns own laptops
- C) Productivity is impossible to measure
- D) Interns dislike technology

Q48 A, B and C can complete a job in 12, 18 and 36 days respectively. How long will they take together?

Correct: B

- A) 5 days
- B) 6 days
- C) 7 days
- D) 8 days

Q49 A family has six members P, Q, R, S, T and U. P is the mother of Q. Q is the brother of R. S is the father of P. T is the sister of P. U is the son of R. How is S related to U?

Correct: B

- A) Father
- B) Grandfather
- C) Great-grandfather
- D) Uncle

Q50 In a row of 20 people, K is 7th from the left. What is K's rank from the right?

Correct: C

- A) 12th
- B) 13th
- C) 14th
- D) 15th

Section 3 — Verbal Ability

20 questions • Correct option is displayed on the right of each question

Q51 Choose the correct verb: The panel of judges ____ decided the winner.	Correct: B
A) have	
B) has	
C) are	
D) were	
Q52 Choose the correct verb: The manager, along with his assistants, ____ attending the meeting.	Correct: C
A) are	
B) were	
C) is	
D) have	
Q53 Choose the correct sentence.	Correct: C
A) Ram was best student.	
B) Ram was a best student.	
C) Ram was the best student.	
D) Ram was an best student.	
Q54 Convert to indirect speech: He said, 'I am busy now.'	Correct: B
A) He said that he is busy now.	
B) He said that he was busy then.	
C) He said that I was busy then.	
D) He says that he was busy now.	
Q55 Choose the correct indirect speech for: The teacher said, 'The Earth revolves around the Sun.'	Correct: B
A) The teacher said that the Earth revolved around the Sun.	
B) The teacher said that the Earth revolves around the Sun.	
C) The teacher says that the Earth revolved around the Sun.	
D) The teacher said that the Earth had revolved around the Sun.	
Q56 Choose the antonym of 'Chastise'.	Correct: C
A) Punish	
B) Scold	
C) Praise	
D) Blame	
Q57 Choose the synonym of 'Hostility'.	Correct: B
A) Friendship	
B) Enmity	
C) Calmness	

D) Kindness

Q58 Choose the closest meaning of 'Sauntering'.

Correct: B

A) Running

B) Strolling

C) Jumping

D) Fighting

Q59 Choose the synonym of 'Omen'.

Correct: A

A) Augury

B) Obstacle

C) Argument

D) Expense

Q60 Choose the correct article: ___ elephant is a vegetarian animal.

Correct: B

A) A

B) An

C) The

D) No article

Q61 Choose the correct preposition: She is good ___ mathematics.

Correct: B

A) in

B) at

C) on

D) for

Q62 Identify the error: 'Each of the students have submitted the form.'

Correct: C

A) Each

B) of the students

C) have

D) the form

Q63 Choose the grammatically correct sentence.

Correct: B

A) He is senior than me.

B) He is senior to me.

C) He is more senior than to me.

D) He senior to me.

Q64 One who can use both hands equally well is called:

Correct: A

A) Ambidextrous

B) Ambiguous

C) Amateur

D) Anonymous

Q65 Choose the best completion: If I had known about the delay, I ___ earlier.	Correct: C
A) will leave	
B) would leave	
C) would have left	
D) left	

Q66 Choose the passive voice: 'The team completed the project.'	Correct: B
A) The project completed the team.	
B) The project was completed by the team.	
C) The team was completed by the project.	
D) The project is completing by the team.	

Q67 Choose the correct word: The new policy will ___ employee morale.	Correct: B
A) effect	
B) affect	
C) effects	
D) affects	

Q68 Choose the correct sentence.	Correct: B
A) Neither the manager nor the employees is ready.	
B) Neither the manager nor the employees are ready.	
C) Neither manager nor employees is ready.	
D) Neither the manager or the employees are ready.	

Q69 Choose the word closest in meaning to 'Vagary'.	Correct: A
A) Impulse	
B) Certainty	
C) Routine	
D) Agreement	

Q70 Read: 'The company introduced flexible hours to help employees manage commuting and family responsibilities.' What is the main purpose of flexible hours?	Correct: B
A) Increase office rent	
B) Support employees' schedules	
C) Reduce salaries	
D) Eliminate meetings	

Section 4 — Technical MCQ & CS Fundamentals

30 questions • Correct option is displayed on the right of each question

Q71 Which SQL command is used to retrieve data from a table?	Correct: B
A) INSERT	
B) SELECT	
C) UPDATE	
D) DELETE	
Q72 Which SQL command removes all rows while retaining the table structure?	Correct: C
A) DROP	
B) DELETE	
C) TRUNCATE	
D) ALTER	
Q73 Which clause filters rows before grouping?	Correct: C
A) HAVING	
B) ORDER BY	
C) WHERE	
D) GROUP BY	
Q74 Which clause filters groups after aggregation?	Correct: B
A) WHERE	
B) HAVING	
C) ORDER BY	
D) FROM	
Q75 Which condition correctly checks for NULL values?	Correct: C
A) salary = NULL	
B) salary == NULL	
C) salary IS NULL	
D) salary EQUAL NULL	
Q76 Which query returns the second-highest distinct salary?	Correct: B
A) SELECT MAX(salary) FROM Employees;	
B) SELECT MAX(salary) FROM Employees WHERE salary < (SELECT MAX(salary) FROM Employees);	
C) SELECT MIN(salary) FROM Employees;	
D) SELECT salary FROM Employees ORDER BY salary;	
Q77 Which pattern finds names beginning with 'A'?	Correct: B
A) LIKE '%A'	
B) LIKE 'A%'	
C) LIKE '_A'	
D) = 'A%'	

Q78 What does GROUP BY department with COUNT(*) return?	Correct: B
A) Total salary per department	
B) Number of employees per department	
C) Only the IT department	
D) Employee names only	
Q79 Which JOIN returns only rows having matching values in both tables?	Correct: D
A) LEFT JOIN	
B) RIGHT JOIN	
C) FULL OUTER JOIN	
D) INNER JOIN	
Q80 Which normal form primarily removes partial dependency on part of a composite key?	Correct: B
A) 1NF	
B) 2NF	
C) 3NF	
D) BCNF	
Q81 Which is a valid primary-key definition?	Correct: A
A) ID INT PRIMARY KEY	
B) ID PRIMARY INT	
C) PRIMARY ID INT	
D) ID INT KEY ONLY	
Q82 What is the main purpose of a foreign key?	Correct: B
A) Ensure every value is unique	
B) Maintain referential integrity between tables	
C) Sort a table	
D) Encrypt a column	
Q83 Which OOP concept bundles data and methods and restricts direct access to some internal state?	Correct: B
A) Inheritance	
B) Encapsulation	
C) Polymorphism	
D) Compilation	
Q84 Which OOP concept exposes essential behavior while hiding implementation details?	Correct: A
A) Abstraction	
B) Inheritance	
C) Overloading	
D) Iteration	

Q85 When the same interface/reference can invoke different overridden implementations at runtime, this is:	Correct: B
A) Encapsulation	
B) Runtime polymorphism	
C) Compilation	
D) Aggregation only	
Q86 Which OOP feature allows a class to acquire properties and methods of another class?	Correct: A
A) Inheritance	
B) Parsing	
C) Normalization	
D) Indexing	
Q87 Can an abstract class be instantiated directly with new in Java?	Correct: C
A) Yes, always	
B) Yes, inside its own class only	
C) No	
D) Only if it has a constructor	
Q88 In Java, what does `String s1="test"; String s2="test"; s1 == s2` evaluate to?	Correct: A
A) true	
B) false	
C) Compilation error	
D) Runtime exception	
Q89 What is the result of storing a Square into an actual Circle[] array in Java?	Correct: B
A) It prints normally	
B) ArrayStoreException at runtime	
C) Compilation always succeeds and prints 3	
D) NullPointerException	
Q90 Which statement about method overriding is correct?	Correct: B
A) It occurs only within one class	
B) A subclass provides a new implementation of an inherited method with compatible signature	
C) It requires different parameter lists	
D) It is the same as constructor overloading	
Q91 What is the maximum number of swaps Selection Sort performs on n elements?	Correct: B
A) n	
B) n-1	
C) n ²	

D) $n/2$

Q92 What is the guaranteed time complexity of Merge Sort?

Correct: C

A) $O(n)$

B) $O(\log n)$

C) $O(n \log n)$

D) $O(n^2)$

Q93 What is the worst-case time complexity of standard Quick Sort with consistently poor pivots?

Correct: D

A) $O(n)$

B) $O(\log n)$

C) $O(n \log n)$

D) $O(n^2)$

Q94 Which sorting algorithm can run in $O(n)$ in the best case on an already sorted array?

Correct: B

A) Selection Sort

B) Insertion Sort

C) Merge Sort

D) Heap Sort

Q95 Which data structure follows LIFO?

Correct: B

A) Queue

B) Stack

C) Heap only

D) Graph

Q96 Which data structure follows FIFO?

Correct: B

A) Stack

B) Queue

C) Tree

D) Set

Q97 What is the key difference between a process and a thread?

Correct: B

A) A process is always smaller than a thread

B) Threads within a process share its address space, while processes have separate address spaces

C) Processes cannot execute concurrently

D) Threads always have separate memory spaces

Q98 Which condition is necessary for deadlock?

Correct: A

A) Mutual exclusion

B) Infinite CPU speed

C) No shared resources

D) Guaranteed preemption

Q99 Which protocol is connection-oriented and provides reliable byte-stream delivery?	Correct: C
A) UDP	
B) IP	
C) TCP	
D) DNS	

Q100 What is the primary purpose of DNS?	Correct: B
A) Encrypt HTTP traffic	
B) Translate domain names to IP addresses	
C) Transfer files	
D) Assign MAC addresses	

Online Preparation Resources

Public sources report that Mphasis assessment patterns can vary by drive. Use these resources for practice, but follow the exact assessment instructions in your invitation/college placement notice.

Mphasis official careers

<https://careers.mphasis.com/home.html>

Use this to verify official openings and recruitment information.

Mphasis official application process

<https://careers.mphasis.com/home/information-center/application-process.html>

Official overview of the recruitment/application process.

CodeStudio — Mphasis Placement Preparation

<https://codestudio.pro/company-prep/mphasis>

Practice-oriented coverage of aptitude, reasoning, verbal, programming MCQs, SQL/OOP/OS and interview preparation.

PlacementPreparation.io — Mphasis

<https://www.placementpreparation.io/mphasis/interview-questions/>

Sample technical and interview questions covering programming, OOP, DBMS/SQL, data structures and project discussion.

PapersAdda — Mphasis Placement Papers 2026

<https://papersadda.com/article/mphasis-placement-papers-2026/>

Recent public placement-paper discussion; useful for seeing how reported patterns vary.

Mphasis placement paper practice archive

<https://placementpapers.app/mphasis/>

Mixed practice material for aptitude, technical MCQs and coding.

Source & scope note

This is a practice document, not an official Mphasis question paper. The uploaded assessment is a 95-question, 2-hour, four-section mock with 25 Quantitative, 25 Logical, 25 Verbal and 20 Technical questions; this 100-question set expands that structure to include 30 technical/CS-fundamentals questions.